

8.3.4

EE24BTECH11013 - MANIKANTA D

Question:

Sketch the graph of $y = |x + 3|$ and evaluate $\int_{-6}^0 |x + 3| dx$.

Solution:

Integral to calculate,

$$J = \int_{-6}^0 |x + 3| dx \quad (0.1)$$

Using the trapezoidal rule,

$$J = \int_a^b f(x) dx \approx h \left(\frac{1}{2} f(x_0) + f(x_1) + f(x_2) + \cdots + f(x_{n-1}) + \frac{1}{2} f(x_n) \right) \quad (0.2)$$

where $h = \frac{b-a}{n}$, and $x_i = a + ih$.

Let $a = -6$, $b = 0$, and $f(x) = |x + 3|$. Choose $n = 4$, so $h = \frac{0 - (-6)}{4} = \frac{6}{4} = 1.5$. The points are:

$$x_0 = -6, x_1 = -4.5, x_2 = -3, x_3 = -1.5, x_4 = 0 \quad (0.3)$$

The corresponding function values are:

$$f(x_0) = |-6 + 3| = 3, f(x_1) = |-4.5 + 3| = 1.5, \quad (0.4)$$

$$f(x_2) = |-3 + 3| = 0, f(x_3) = |-1.5 + 3| = 1.5, f(x_4) = |0 + 3| = 3 \quad (0.5)$$

Substitute into the trapezoidal rule formula:

$$J \approx 1.5 \left(\frac{1}{2} f(x_0) + f(x_1) + f(x_2) + f(x_3) + \frac{1}{2} f(x_4) \right) \quad (0.6)$$

$$J \approx 1.5 \left(\frac{1}{2} (3) + 1.5 + 0 + 1.5 + \frac{1}{2} (3) \right) \quad (0.7)$$

$$J \approx 1.5 (1.5 + 1.5 + 0 + 1.5 + 1.5) \quad (0.8)$$

$$J \approx 1.5 \times 6 = 9 \quad (0.9)$$

The approximate value of the integral using the trapezoidal rule is $J \approx 9$.

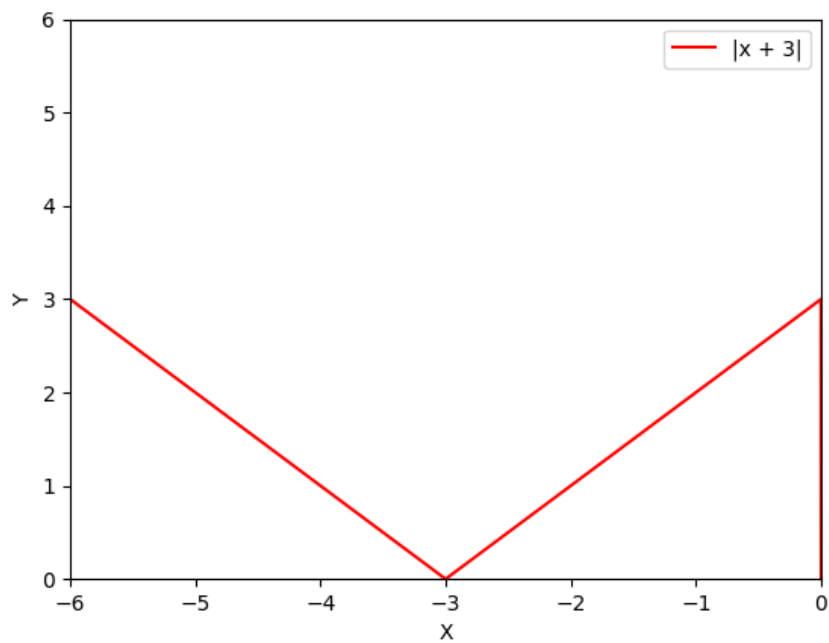


Fig. 0.1: Plot of the differential equation